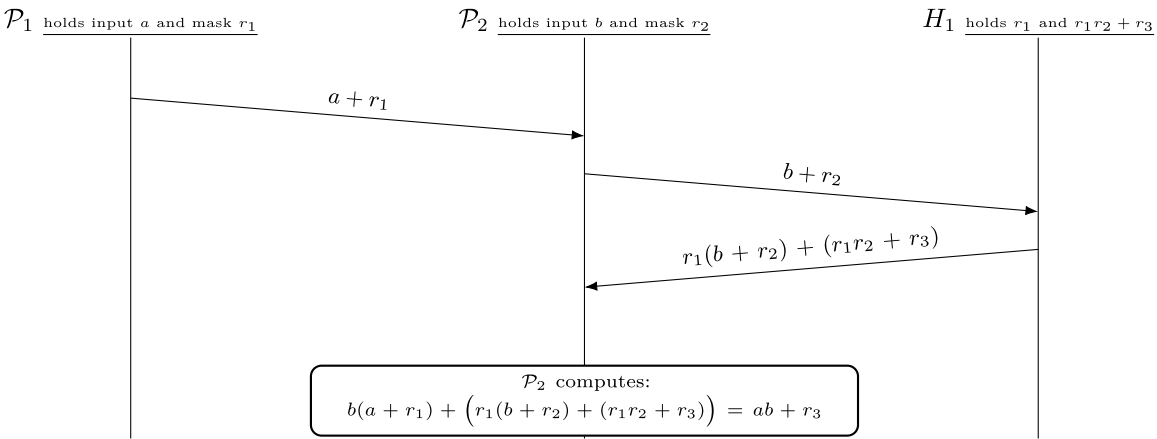
$\mathcal{P}_1$ holds input a and mask r_1
$\mathcal{P}_2$ holds input b and mask r_2
H_1 holds r_1 and $r_1 r_2 + r_3$
$a + r_1$
$b + r_2$
$r_1(b + r_2) + (r_1 r_2 + r_3)$
$\mathcal{P}_2$ computes:
$b(a + r_1) + \Big(r_1(b + r_2) + (r_1 r_2 + r_3)\Big) = ab + r_3$